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Comparison of Bioelectrical Impedance Indices for Skeletal Muscle Mass and Intracellular Water Measurements of Physically Active Young Men and Athletes

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ABSTRACT

Background: Bioelectrical impedance analysis (BIA) is a minimally invasive, safe, easy, and quick technology used to determine body composition.

Objectives: We compared the relationship among impedance indices obtained using single-frequency BIA, multi-frequency BIA, bioelectrical impedance spectroscopy (BIS), and skeletal muscle mass (SMM) of physically active young men and athletes using the creatine (*methyl-d₃*) dilution method. We also compared the SMM and intracellular water (ICW) of athletes and active young men measured using a reference stable isotope dilution and BIS method, respectively.

Methods: We analyzed data from 28 men (mean age, 20 ± 2 y) who exercised regularly. Single-frequency BIA at 5 kHz and 50 kHz (R_5 and R_{50}), multi-frequency BIA (R_{250-5}), and BIS (R_{ICW}) methods of determining the SMM were compared. The deuterium and sodium bromide dilution methods of obtaining the total body water, ICW, and extracellular water measurements were also used, and the results were compared to those acquired using bioimpedance methods.

Results: The correlation coefficients between SMM and L^2/R_5 , L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} were 0.738, 0.762, 0.790, and 0.790, respectively ($P < 0.01$). The correlation coefficients between ICW and L^2/R_5 , L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} were 0.660, 0.687, 0.758, and 0.730, respectively ($P < 0.001$). However, the correlation coefficients of L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} for SMM and ICW were not significantly different.

Conclusions: Our findings suggest that single-frequency BIA at L^2/R_{50} , multi-frequency BIA, and BIS are valid for assessing the SMM of athletes and active young men. Additionally, we confirmed that the SMM and ICW were correlated with single-frequency BIA, multi-frequency BIA, and BIS. Bioimpedance technologies may be dependable and practical means for assessing SMM and hydration compartment status of active young adult males; however, cross-validation is needed.

Keywords: BIA, BIS, skeletal muscle mass, body composition, electrical resistance

Abbreviations: BIA, bioelectrical impedance analysis; BIS, bioelectrical impedance spectroscopy; CCC, concordance correlation coefficient; ECW, extracellular water; ICW, intracellular water; LoA, limit of agreement; R, resistance; SMM, skeletal muscle mass; TBW, total body water; Xc, reactance; Z, impedance.

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Introduction

To maximize performance without compromising health, athletes must assess and monitor energy intake and expenditure throughout the competitive season [1]. Energy requirements and energy balance can be estimated based on body composition and associated changes in body composition [1,2]. Skeletal muscle mass (SMM) assessments are important for identifying the effects of daily training, especially for athletes managing their body weight [3].

SMM can be evaluated using an estimation equation involving appendicular lean soft tissue mass estimated using DXA [2,4]. It can also be assessed using MRI or CT [5]. According to recent validation studies, SMM can also be estimated qualitatively and accurately using the creatine (*methyl-d₃*) dilution (D_3 -creatine) method [6,7]. These measuring devices are often only located in hospitals or research facilities; therefore, more convenient, equally reliable, and relatively quick assessment methods are required for their periodic use during the training cycles of athletes. Bioelectrical impedance analysis (BIA) is inexpensive, noninvasive, easy to operate, portable, and quick; therefore, it can be a practical method of obtaining measurements within 5–10 min [8–10]. However, validation of its usefulness for SMM measurements of athletes and comparisons of its measurements with those obtained using other methods has not been undertaken rigorously.

There are 3 main BIA methods as follows: single-frequency BIA, multi-frequency BIA, and bioelectrical impedance spectroscopy (BIS) [11]. Single-frequency BIA is the most common method—it measures the impedance (Z) or resistance (R) at 50 kHz [12]. With single-frequency BIA, reactance (X_c) is <10% of R, and the magnitudes of R and Z are similar and often used interchangeably; however, the equation is strictly $Z = (R^2 + X_c^2)^{1/2}$ [11]. The frequency of 50 kHz that is currently used for single-frequency BIA does not completely penetrate the tissue; therefore, it may not measure the entire muscle mass or that of other tissue. Multi-frequency BIA can separately evaluate intracellular and extracellular compartments of total body water (TBW) [13]. This is because the cell membrane acts as a capacitor, which makes it possible to measure intracellular water (ICW) and extracellular water (ECW) at different frequencies. Low frequencies such as 5 (R_5) and 50 (R_{50}) kHz primarily measure ECW, whereas high frequencies such as 200–500 kHz measure both ICW and ECW because the cell membrane is permeable to the current [12]. Some studies aimed to develop models for estimating SMM based on bioelectrical impedance. During those studies, single-frequency BIA was compared with MRI. The bioelectrical impedance index, which is calculated as the square of the segment length (L) divided by Z_{50} or R_{50} , was found to be strongly correlated with SMM. BIS characterizes the measurement section with a parallel circuit of ECW and ICW and uses a series of frequencies according to the principles of the Cole model. By mathematically fitting the measured raw data to this model, R at an infinite frequency (R_∞) and R at 0 frequency (R_0) were obtained by extrapolation [13,14].

Most studies that attempted to develop R or Z models for estimating SMM have used single-frequency BIA instead of MRI. These studies found that the bioelectric Z index (L^2/Z_{50}), which is the square of the length (L) divided by Z (Z_{50}) or R (R_{50}) at 50 kHz, is strongly correlated with SMM; however, the available validation

studies involved healthy young adults [15,16]. Skeletal muscle quality varies with the collagen concentration, elastic fiber system, fat accumulation in skeletal muscles, and expansion of ECW relative to the skeletal muscle [18]. Changes in these parameters affect the relationships among single-frequency BIA, multi-frequency BIA, BIS, and SMM. Compared to healthy adults without high levels of training, athletes may be a more homogeneous population with less inter-muscle or intra-muscle fat accumulation, potentially more connective tissue, more intra-muscle protein, and more ICW. This means that the aforementioned correlations between impedance measures might not be observed in this population, and it is important to understand these differences. Therefore, it is necessary to verify the relevance and validity of each frequency (impedance index) for athletes using substantial SMM and accurate TBW, ICW, and ECW as references (hereafter referred to as SMM, TBW_{REF} , ICW_{REF} , and ECW_{REF} , respectively).

This study aimed to compare the relationship among impedance indices obtained using single-frequency BIA, multi-frequency BIA, and BIS and whole-body SMM of physically active young men and athletes determined using the D_3 -creatine dilution method. We also compared the SMM and ICW of athletes and active young men determined using a reference stable isotope dilution and BIS method, respectively.

Methods

Ethical considerations

Participants were invited to attend an informational meeting at which they were informed of the benefits and potential risks of the investigation. All the Japanese participants provided written informed consent. The Institutional Review Boards of the Japan Institute of Sports Sciences (Approval No.: 050) and the Faculty of Health and Sport Sciences, University of Tsukuba (Approval No.: 019-33) approved the protocol. This study was performed in accordance with the ethical guidelines of the Declaration of Helsinki.

Study design and protocol

This cross-sectional trial was performed at a single site (Japan Institute of Sports Science, Tokyo, Japan). A total of 31 healthy, active, young male adults older than 18 y of age were enrolled through prospective open recruitment. Participants were members of a college sports club or an athletic organization (judo, triathlon, wrestling, orienteering, and cycling) or had participated in recreational sports and/or performed athletic training during weekends. They were divided into the following 3 groups for analysis: endurance group (triathlon, orienteering, and cycling; $n = 13$); combat group (judo and wrestling; $n = 8$); and active young group ($n = 7$). Participants were included if they did not use any medications and creatine supplements, had no history of thyroid and heart diseases, and had no extreme dietary restrictions. Three individuals were excluded from the study; 1 was excluded because of missing BIS data, and 2 were excluded because of missing D_3 -creatine data. Ultimately, 28 participants were included in the final analyses (age, 20 ± 2 y; height, 171.1 ± 5.9 cm; body mass, 68.7 ± 9.8 kg). The protocol and methodology of the D_3 -creatine dilution method used for SMM measurements have been described [19,20]. The day prior to the evaluation of body composition, the participants resided at our research facility and refrained from engaging in high-intensity

exercise for 24 h. They were provided dinner at the laboratory and subsequently asked to fast for ≥ 10 h.

TBW, ICW, and ECW

The deuterium dilution technique was used to determine TBW_{REF}. Each participant consumed ~ 0.12 g/kg of $^2\text{H}_2\text{O}$ (99.8%; Taiyo Nippon Sanso). The actual dose was measured using an electric scale (± 0.1 mg; XPE205; Mettler Toledo). Urine samples were collected at baseline, 3 h, and 4 h. The baseline and 4 h urine specimens were evaluated to determine the TBW_{REF}. The urine samples were frozen for later analyses using isotope ratio MS (Hydra 20-20; Sercon Ltd.). The detailed protocol used for isotope ratio MS analysis has been described previously [21].

The ECW_{REF} was determined using the sodium bromide (NaBr) dilution method [22,23]. NaBr was included with the $^2\text{H}_2\text{O}$ dose so that the individuals received 10 mg NaBr/kg of body mass. Plasma samples were obtained from the blood samples collected at baseline, 3 h, and 4 h and frozen for later analyses using an inductively coupled plasma mass spectrometer (ELEMENT XR; RF Power 1200 W; ThermoFisher Scientific). These specimens were analyzed at Toray Research Center Inc. (Tokyo, Japan). ECW_{REF} was calculated as follows:

$$\text{ECW}_{\text{REF}} = \{\text{Br dose} / (\text{Br}_{4\text{h}} - \text{Br}_{0\text{h}})\} \times 0.90 \times 0.95$$

where bromide (Br) denotes the plasma Br concentration, and the constants (0.90 and 0.95) are used to adjust for the over-expansion of Br into non-extracellular sites and for the Donnan equilibrium effect. ICW_{REF} was computed as follows:

$$\text{ICW}_{\text{REF}} = \text{TBW}_{\text{REF}} - \text{ECW}_{\text{REF}}$$

Multi-frequency BIA and BIS

The individual's height was used as the length (L) in the equations. Impedance measurements were obtained with the participants in a relaxed supine position on a padded bed, with the arms slightly abducted from the body, forearms pronated, and legs slightly apart [24]. Participants were instructed to refrain from vigorous exercise for 24 h and to fast on the experiment day and on the day when other body composition measurements were performed. The room temperature was set to maintain a comfortable environment (21°C–22°C). Bioelectrical impedance was measured after 15 min of rest to avoid the immediate (1–2 min) effect of the transition from the standing position to the supine position on the shift in body fluid from the extremities to the thorax as well as the slow phase of this shift that continues for up to 3–12 h [24]. Bioelectrical impedance was measured using a logarithmic distribution of 256 frequencies, ranging from 4 to 1000 kHz (SFB7; ImpediMed) with disposable tab-type monitoring electrodes (2 cm $\times$ 2 cm; Red Dot; 3M). All the measurements were obtained through a visual examination of the raw data (the Td correction was turned off) while focusing on identifying a clear semicircle. The goal was to achieve a SEE of <1.0 , and all Cole plots had a SEE of <1.0 (Table 1); therefore, the Td correction was unnecessary. Before all the measurements were performed, the device was evaluated based on the series of precision resistors of the manufacturer [25].

The R_0 and R_∞ were determined by extrapolation after fitting the spectrum of bioimpedance data to the Cole model [14] using

TABLE 1

Bioimpedance indices and comparisons of total body water, intracellular water, and extracellular water

	Mean		SD
Height (cm)	171.1	$\pm$	5.9
R_5 (Ω)	565.6	$\pm$	64.3
L^2/R_5 (cm^2/Ω)	52.4	$\pm$	6.8
Z_5 (Ω)	566.7	$\pm$	64.4
L^2/Z_5 (cm^2/Ω)	52.3	$\pm$	6.8
R_{50} (Ω)	470.3	$\pm$	54.8
L^2/R_{50} (cm^2/Ω)	63.1	$\pm$	8.4
Z_{50} (Ω)	474.5	$\pm$	55.3
L^2/Z_{50} (cm^2/Ω)	62.6	$\pm$	8.3
R_{250-5} (Ω)	1512.7	$\pm$	205.7
L^2/R_{250-5} (cm^2/Ω)	19.7	$\pm$	3.0
Z_{250-5} (Ω)	1523.3	$\pm$	208.8
L^2/Z_{250-5} (cm^2/Ω)	19.6	$\pm$	3.0
R_0 (Ω)	594.5	$\pm$	67.3
R_∞ (Ω)	381.4	$\pm$	45.1
R_{ICW} (Ω)	1067.6	$\pm$	150.4
L^2/R_{ICW} (cm^2/Ω)	28.0	$\pm$	4.4
SEE of Cole plot (%)	0.41	$\pm$	0.13
TBW _{REF} (kg)	43.3	$\pm$	4.8 ²
TBW _{BIS} (kg)	42.9	$\pm$	5.8
ICW _{REF} (kg)	24.0	$\pm$	3.1
ICW _{BIS} (kg)	25.2	$\pm$	3.6 ^{1,2}
ECW _{REF} (kg)	19.3	$\pm$	2.7
ECW _{BIS} (kg)	17.7	$\pm$	2.3 ¹
SMM (kg)	31.6	$\pm$	5.7
eSMM _{BIA} (kg)	32.8	$\pm$	3.4

Values are means and SD.

Abbreviations: BIA; bioelectrical impedance analysis; BIS, bioimpedance spectroscopy; ECW, extracellular water; eSMM; estimated skeletal muscle mass; ICW, intracellular water; REF, reference method; SMM, skeletal muscle mass; TBW, total body water.

¹ $P < 0.01$ BIS compared with the reference method.

² $P < 0.01$ ICW and TBW compared with SMM.

specific software (Bioimp version 5.5.0.1; Impedimed) based on original methods derived by Cornish et al. [26]. The analysis parameters were as follows: minimum frequency, 5 kHz; maximum frequency, 500 kHz; and rejection limit, 0%. Data regarding 256 frequencies were obtained from BIS, and R_5 , R_{50} , and R_{250-5} values were extracted. For BIS, R_{ICW} was calculated using $1/[(1/R_\infty) - (1/R_0)]$ and for multi-frequency BIA, R_{250-5} was computed using $1/[(1/R_{250}) - (1/R_5)]$. The impedance indices were obtained from the L^2/R_5 , L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} formulas for the whole body [11]; thereafter, the specific measured frequency values of R_5 , R_{50} , and R_{250-5} were extracted to emulate single-frequency and multi-frequency BIA. The TBW_{BIS}, ICW_{BIS}, and ECW_{BIS} values were obtained using the software (Bioimp version 5.5.0.1; Impedimed). Finally, we used the default calculation system of SFB7, which is based on De Lorenzo's equation, and used it for the TBW_{BIS} calculation [27]. We performed calculations using BIA at 50 kHz to determine the estimated SMM (eSMM_{BIA}) based on the previous equation: $\text{eSMM}_{\text{BIA}} = [(\text{height}^2/R \times 0.401) + (\text{sex} \times 3.825) + (\text{age} \times -0.071)] + 5.102$, where height is in centimeters, R is in ohms, and age is in years (regarding sex, men = 1 and women = 0) [17].

Statistical analysis

The results are presented as the mean $\pm$ SD. The TBW, ICW, and ECW data were compared with those obtained using the reference method with a paired t-test. A linear regression

analysis was performed to assess the association between the SMM measured using the D₃-creatine, ICW, and impedance indices. The r were compared using the Meng–Rosenthal–Rubin method z -test [28]. The Bland–Altman plot analysis was performed to compare different methods and to determine the 95% limit of agreement (LoA) [29]. The regression-based 95% LoA was used if there were significant relationships between the difference and mean values of the different methods used [30]. A linear regression analysis was performed to investigate the relationship between TBW values estimated using BIS and the reference method. To assess the accuracy of TBW estimated using BIS, Lin's concordance correlation coefficient (CCC) was calculated [31]. The CCC (ρ_c) includes the factor of precision (ρ) and accuracy (Cb), i.e., $\rho_c = \rho \times \text{Cb}$, where ρ is Pearson's r , and Cb is a bias correction factor that measures how far the best-fit line deviates from a line at 45 degrees. Outliers were identified by Cook's distance, with a cut-off value of <0.5 (Figure 1). Statistical analyses were performed, and figures were generated using RStudio software for Mac (RStudio version 1.3.959, 2009–2020 RStudio, Inc.) and Microsoft Excel for Mac version 16.49. For all the analyses, the statistical significance was set at P value <0.05 . Based on the power of 0.80, an effect size of 0.5, and an alpha of 0.05, a total sample size of 26 was necessary for the correlation analysis (G*Power version 3.1.9.6; Universität Kiel).

Results

Table 1 shows the characteristics and bioimpedance (R and Z) indices of the participants. The SEE of curve fitting for the Cole

plot was $<1\%$ (Table 1). TBW_{REF} was not different from TBW_{BIS} . However, there were significant differences in ICW_{REF} and ECW_{REF} compared with ICW_{BIS} and ECW_{BIS} , respectively (Table 1).

Figure 1 shows the relationships between the impedance indices and SMM (Figure 1A–D) and whole-body ICW_{REF} (Figure 1E–H). No outliers were identified by Cook's distance; therefore, all the plots were included (mean 0.048, 0.050, 0.046, 0.047, 0.041, 0.040, 0.031, and 0.035) (Figure 1A–H). All impedance indices were highly correlated with SMM (L^2/R_5 , $r = 0.738$; L^2/R_{50} , $r = 0.762$; L^2/R_{250-5} , $r = 0.790$; and L^2/R_{ICW} , $r = 0.790$) ($P < 0.01$) and ICW_{REF} (L^2/R_5 , $r = 0.660$; L^2/R_{50} , $r = 0.687$; L^2/R_{250-5} , $r = 0.758$; and L^2/R_{ICW} , $r = 0.730$) ($P < 0.01$). The r of SMM tended to be greater for L^2/R_{50} than that for L^2/R_5 ($P = 0.056$). There was no difference in the r of L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} with SMM. The r of ICW_{REF} tended to be greater for L^2/R_{50} than that for L^2/R_5 ($P = 0.059$). There was no difference in the r of L^2/R_{50} , L^2/R_{250-5} , and L^2/R_{ICW} with ICW_{REF} .

The TBW_{REF} measured with deuterium dilutions was strongly correlated with TBW_{BIS} ($r = 0.941$; $P < 0.001$) (Figure 2A). The ICW_{REF} was strongly correlated with ICW_{BIS} ($r = 0.749$; $P < 0.01$) (Figure 2B). The ECW_{REF} was also strongly correlated with ECW_{BIS} ($r = 0.815$; $P < 0.01$) (Figure 2C). The Bland–Altman plot for BIS in relation to the reference method is shown in Figure 2D–F. The mean bias values for TBW, ICW, and ECW were -0.39 kg (95% CI: $-1.21, 0.43$ kg), 1.21 kg (95% CI: $0.26, 2.16$ kg), and -1.60 kg (95% CI: $-2.22, -0.99$ kg), respectively. The differences in TBW_{REF} and TBW_{BIS} were related to the mean ($r = 0.563$; $P < 0.01$; Figure 2D). Thus, we could calculate the regression-based 95% LoA as $-9.45 + 0.21A \pm 1.96 \times 1.81$ kg,

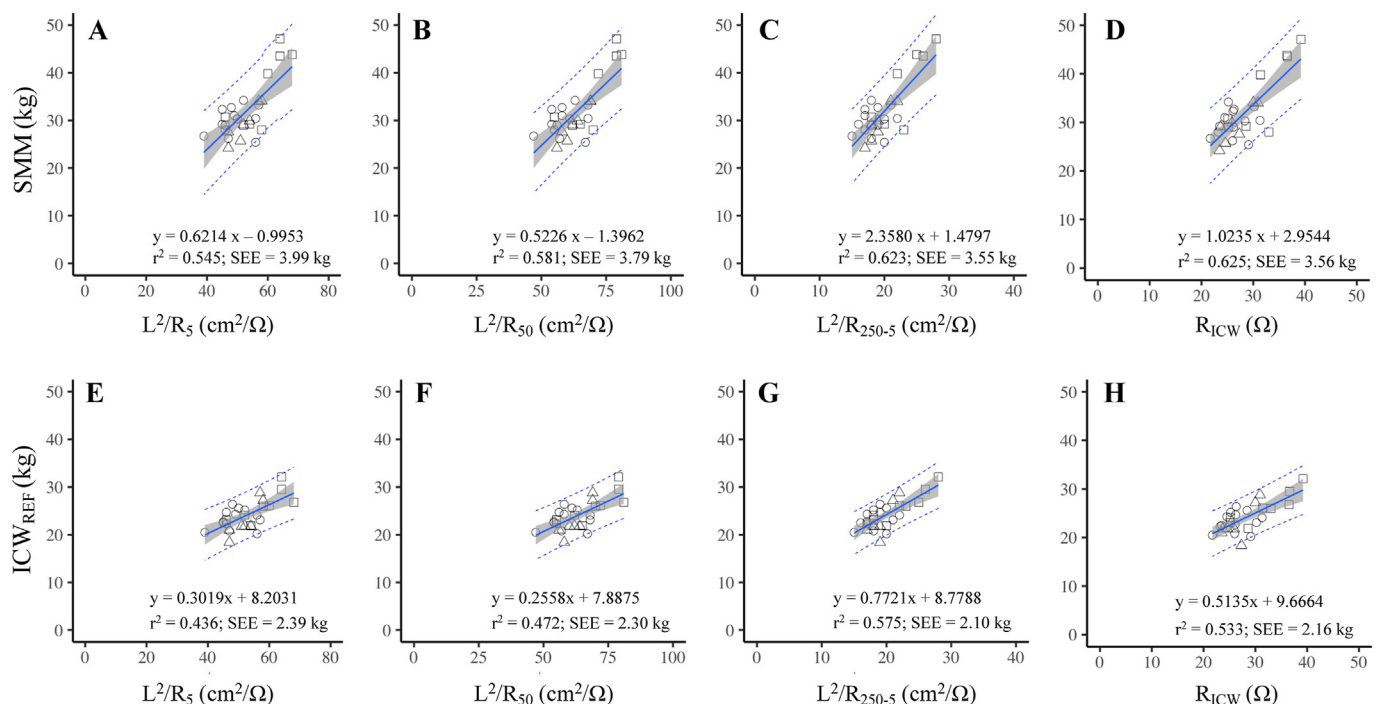


FIGURE 1. Relationship between skeletal muscle mass using D₃-creatine, referenced intracellular water, and impedance indices. Resistance (R) indices of the squared segment length (L^2) divided by R at 5 kHz (A and E), L^2/R_{50} (B and F), L^2/R_{250-5} (C and G), and L^2/R_{ICW} using bioelectrical impedance spectroscopy (BIS) (D and H) in the whole body. The dependent variable in panels A–D represents SMM (kg). The dependent variable in panels E–H represents ICW_{REF} (kg). The gray area represents the 95% CI. Two blue dashed lines represent prediction intervals. SMM, skeletal muscle mass using D₃-creatine dilution, ICW_{REF} , intracellular water using the sodium bromide (NaBr), and stable isotope dilution method. Each plot shows the endurance group (○), combat group (□), and active young group (△). Abbreviations: ICW, intracellular water; SMM, skeletal muscle mass.

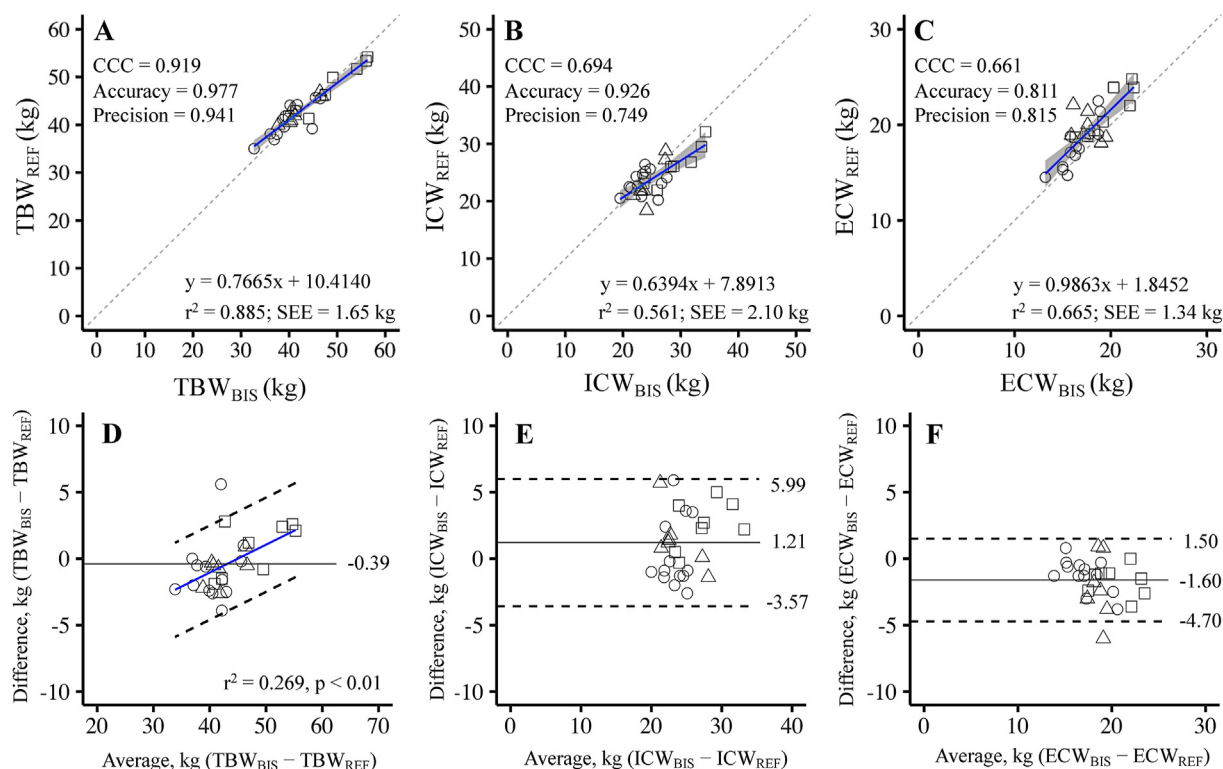


FIGURE 2. Comparison of bioimpedance spectroscopy and the stable isotope method for measuring total body water (TBW), extracellular water (ECW), and intracellular water (ICW). Correlations between the methodologies used to measure TBW (A), ICW (B), and ECW (C) were assessed. The gray area represents the 95% CI. The solid bold line represents the regression line. The dashed line represents the identity line in panels A–C. Bland–Altman agreement plots of the differences among the methods used to measure TBW (D), ICW (E), and ECW (F) are shown here. The solid line represents the mean in panels D–F. The dashed line is represented based on the regression line ± 1.96 residual SD in panel D. The dashed line represents the mean ± 1.96 SD in panels E and F. Each plot shows the endurance group ($\circ$), combat group ($\square$), and active young group ($\triangle$). Abbreviation: CCC, concordance correlation coefficient.

where A is the mean of TBW_{REF} and TBW_{BIS} (Figure 2D). The 95% LoA for TBW, ICW, and ECW were -5.89 to 5.70 kg (Figure 2D), -3.57 to 5.99 kg (Figure 2E), and -4.70 to 1.50 kg (Figure 2F), respectively. There was no trend in the differences between ICW_{REF} and ICW_{BIS} (Figure 2E) or in the differences between ECW_{REF} and ECW_{BIS} (Figure 2F). Figure 3 shows the correlations between SMM and ICW_{REF} ($r = 0.822$; $P < 0.01$;

Figure 3A) and between SMM and ICW_{BIS} ($r = 0.819$; $P < 0.01$; Figure 3B). The intercept was not significantly different from 0 (95% CI: -15.02 , 5.46); however, the slope was significantly higher than 1.00 (95% CI: 1.09 , 1.94) (Figure 3A). The intercept of the regression line did not differ significantly from 1.00 (95% CI: -10.17 , 8.36), and the slope did not differ significantly from 0 (95% CI: 0.92 , 1.65) (Figure 3B). There was a correlation

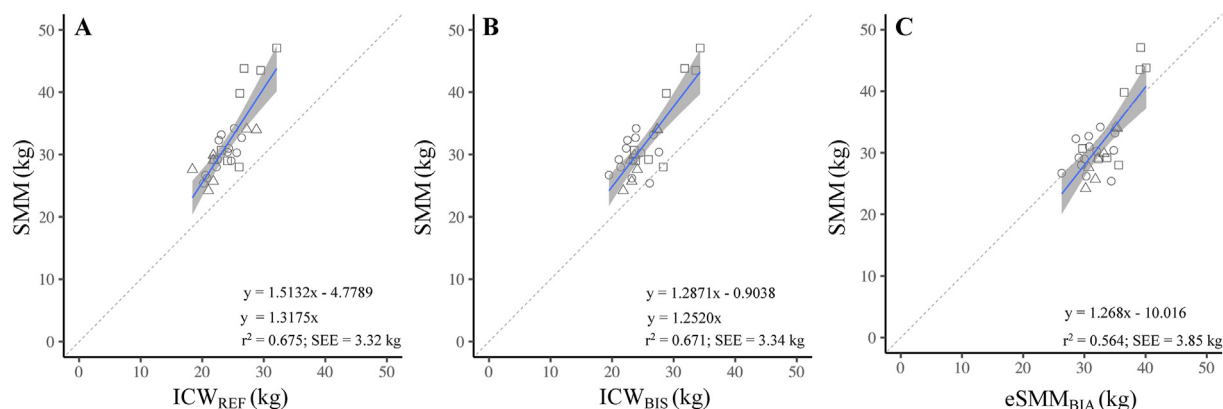


FIGURE 3. Relationship between skeletal muscle mass and intracellular water measured using different methods. (A) Correlation between SMM and ICW_{REF}. (B) Correlation between SMM and ICW_{BIS}. The gray area represents the 95% CI. The solid bold line represents the regression line. The dashed line represents the identity line. SMM measurements using D₃-creatine dilution, ICW_{REF}, and intracellular water using the sodium bromide (NaBr) and stable isotope dilution methods are shown. Each plot shows the endurance group ($\circ$), combat group ($\square$), and active young group ($\triangle$). Abbreviations: ICW, intracellular water; SMM, skeletal muscle mass.

between SMM and eSMM_{BIA} ($r = 0.751$; $P < 0.01$) (Figure 3C), and no difference was observed between the measured SMM and eSMM_{BIA} (31.6 ± 5.7 and 32.8 ± 3.4 ; $P = 0.102$) (Table 1). The r of SMM tended to be greater for ICW_{BIS} than that for eSMM_{BIA} ($P = 0.061$) (Figure 3).

Discussion

The main findings of the study were that SMM is significantly highly correlated with single-frequency BIA (L^2/R_5 and L^2/R_{50}), multi-frequency BIA (L^2/R_{250-5}), and BIS (L^2/R_{ICW}). The r of single-frequency BIA at L^2/R_{50} with SMM tended to be larger than those of single-frequency BIA at L^2/R_5 for athletes and active young men.

The bioimpedance technique does not directly measure body composition; instead, it measures bioelectrical resistance. Several assumptions are required to derive physiological information [32,33]. The 2 main parameters measured by the bioimpedance tool are R and X_c . The estimation equations for SMM, FFM, and body fat from the measured R , X_c , and impedance are derived from regression calculations. The results are obtained using other secondary indirect body composition measurement methods, such as MRI, underwater weighing, or DXA. Therefore, some multiple regression equations have shown that additional factors such as height, weight, sex, and age can attenuate estimation variations [34]. These estimation equations are adaptable to individuals with characteristics similar to those of the population for which they were developed; however, they may introduce additional estimation errors for individuals with significantly different physiological body factors, such as athletes and physically active men [10,35]. We examined the potential for impedance measurements to estimate SMM based on reasonably reliable r^2 values associated with SMM and impedance indices of active young men, including athletes, using the D₃-creatine method. We have previously shown the use of D₃-creatine for measuring SMM against actual SMM measurement methods [20]. The present study is the first to compare impedance indices obtained with the SMM criteria method for this specific population. These impedance indices ranging from 54.5% to 62.5% of the variation in SMM were determined by BIA or BIS. In fact, because impedance does not provide SMM values in kilograms, it is not possible to determine how far apart or close the values of SMM measured by impedance are to those obtained using the isotope method. However, it does indicate the strength of the association between the 2 methods. In our study, we used height as a proxy for length (L); however, we speculated that the athletes' arm and leg lengths and thicknesses might vary between individuals who are proficient in different types of sports, even if they have the same height, which can affect the SMM variations. A previous study validated the impedance indices (50 kHz BIA) using MRI for muscle assessments among a population with diverse body compositions [36]. Furthermore, for healthy young men, the L^2/Z_{50} values of the forearm, upper arm, lower leg, and thigh muscle volume obtained using segmented single-frequency BIA strongly correlated with the muscle volume of all body segments observed using MRI ($r > 0.90$) [37]. However, the relationships between multiple impedance indices for the whole body of athletes have not been fully elucidated. Our whole-body SMM results also showed moderate r with impedance indices ($r > 0.70$) for all single-frequency and

multi-frequency BIA and BIS methods. A unique previous study that included participants 20 to 81 y of age and compared SMM measured using the D₃-creatine method with BIS showed correlations in SMM and FFM ($r = 0.8857$) [19]. Our study found a stronger correlation between SMM and FFM ($r = 0.9407$) converted from TBW/0.732 (Figure 2A), which may be attributable to our study being limited to young, active men. It should be noted that our results were obtained from a participant group including athletes with relatively large SMM. Nevertheless, we were able to demonstrate a relationship between resistance and SMM using single-frequency BIA, in addition to BIS and multi-frequency BIA.

The ICW and ECW can be calculated from raw bioimpedance data and used to determine the compartmental hydration status [38]. Water, comprising a large amount of SMM (~79%) [39], is also an important indicator of muscle quality in athletes and has roles in muscle strength, upper body power, maximum forearm strength, leg strength, and jump height [11,40–42]. Thus, raw bioimpedance and hydration data are meaningful to athletes and clinicians. Therefore, in addition to the frequently used single-frequency BIA at R_{50} , we used the multi-frequency approach, which is capable of distinguishing between intracellular and extracellular cells through the amalgamation of R_{250-5} , and the BIS method, which indirectly evaluates muscle mass as a parameter deduced by arbitrary algorithms or utilizing parameters such as R_0 and R_∞ derived from the Cole analysis [14]. The ICW results validated the 4 types of impedance indices observed using the isotope dilution technique. There were no differences in the r for single-frequency BIA at L^2/R_{50} , multi-frequency BIA (L^2/R_{250-5}), and BIS (L^2/R_{ICW}) for the impedance indices in this study, suggesting that ICW can be similarly estimated using these impedance indices. The r was slightly lower at 5 kHz than that at 50 kHz. Low frequencies, typically 1 kHz or 5 kHz, are used to estimate ECW because of the negligible penetration of ICW at these frequencies [43]. At 5 kHz, the signal pathway is almost exclusively extracellular, leading to a linear relationship between the R index and ECW. We speculate that this explains the tendency for SMM to be lower at 5 kHz than at 50 kHz because this reflects the relationship between ECW and SMM. Therefore, 50 kHz is better when using single frequencies because low frequencies do not penetrate cells or SMM. Interestingly, R_5 had a moderate correlation and is commonly used with R_{250} as an indicator during multi-frequency BIA and in BIS as an indicator of ECW components based on low-frequency characteristics during BIS. We further compared the water assessments results obtained using BIS and the original software (Bioimp version 5.5.0.1) with the reference method, especially for athletes, because of the higher-accuracy TBW, ICW, and ECW estimations than those obtained using BIA during a previous study [44]. ICW_{BIS} was overestimated (bias, 1.2 kg), ECW_{BIS} was underestimated (bias, -1.6 kg), and TBW_{BIS} was similar to those observed with the reference method (Table 1 and Figure 2). The bias for TBW was 0.4 kg, which is comparable to the bias obtained by previous studies involving wrestlers (0.3 kg) [25]. This bias is smaller than the measurement error of 0.5 kg for TBW shown by our previous study [45], and it is considered clinically negligible. Furthermore, a proportional bias was found (Figure 2D), as well as our previous study in wrestlers using the same BIS system [25]. However, the LoA obtained from the Bland–Altman plot were -5.89 to 5.70 kg for TBW, -3.57 to

5.99 kg for ICW, and -4.70 to 1.50 kg for ECW. These LoA are slightly larger than those reported in the studies by Shiose et al. [25] and Francisco et al. [38]. When the mean ICW_{REF} was 24 kg, LoA was ± 4.8 kg, which amounts to a difference of 20% (Figure 2E). This relative disagreement would be large for some athletes in this study and a previous study [46]; therefore, BIS provides an accurate measurement compared to the reference method. However, its use for assessing water compartments at the individual concentration should be further considered because of the large relative disagreement. Although we attempted to measure these indices while the athletes were hydrated, it is possible that different dehydration levels and sports proficiencies may have affected the magnitude of these LoA. Specifically, anthropometric characteristics that are unique to athletes may be related to the LoA. For example, although anthropometry is incorporated into the equation for estimating body water through the variable Kb [46], the equation was not developed specifically for Japanese individuals or athletes. Regarding athletes, the discrepancy between the actual and assumed Kb values may have been greater, resulting in inaccurate body water estimation (Figure 2D–F). Moreover, R_0 and R_∞ obtained from extrapolation of the curvilinear regression are likely to have more significant estimated errors because measuring reactance at low and high frequencies is challenging [11]. The results are similar to those of previous studies that also validated using the dilution method among a healthy general population [22,25,47,48]. They are also consistent with those of our previous study of the validity of TBW_{BIS} obtained using the same technique for wrestlers [25]. Regarding the analysis of r , Armstrong et al. [49] tested the BIS and isotope dilution techniques among 30 healthy college male students and reported similar coefficients of determination for TBW , ECW , and ICW . Additionally, Francisco et al. [38] performed attempted to validate the use of the isotope dilution method for athletes and showed that BIS more effectively estimated TBW ($r^2 = 0.98$; $SEE = 1.44$; $CCC = 0.99$), with higher coefficients of determination and lower estimated errors than ICW ($r^2 = 0.91$; $SEE = 1.99$; $CCC = 0.95$) in well-hydrated athletes. Similarly, our findings demonstrated higher correlations and reduced estimation errors for TBW ($r^2 = 0.86$; $SEE = 1.65$; $CCC = 0.92$), thus confirming their validity. Nevertheless, as evidenced by our previous investigation, several computational processes are involved in ICW measurements, and further investigations are required [25]. Consequently, BIS may be a good instrument for assessing TBW at the group level among athletes compared to reference methods; however, BIS is limited at the individual level.

We identified that there are associations between ICW and SMM , which are not unexpected considering that most ICW is contained in muscle cell mass. Considering that this association was observed for both ICW_{REF} and ICW_{BIS} , it could also help estimate SMM using BIS and the stable isotope dilution technique. Furthermore, it is possible to approximately estimate SMM by multiplying ICW_{REF} by 1.3175 (95% CI: 1.264 , 1.371 ; Figure 3A) and ICW_{BIS} by 1.2520 (95% CI: 1.202 , 1.302 ; Figure 3B). When estimating SMM by using these coefficients and multiplying ICW_{REF} or ICW_{BIS} , we found no differences in SMM estimated from ICW_{REF} (31.6 ± 4.1 kg; $P = 0.11$) and from ICW_{BIS} (31.6 ± 4.6 kg; $P = 0.16$), compared with SMM measured by the MRI method, as described by our previous validation

study (32.3 ± 3.9 kg), respectively [2]. Consequently, SMM could be estimated using these coefficients for ICW with BIS or the stable isotope dilution method. Additionally, we compared the measured SMM and estimated SMM based on a previously established equation. No significant disparity was observed between the measured SMM and $eSMM_{BIA}$ (31.6 ± 5.7 and 32.8 ± 3.4 ; $P = 0.102$). This finding lends support to the utilization of bioelectrical impedance indices because these predictors can be used for estimating SMM and encompasses variables such as resistance, height, age, and sex. Moreover, the r of SMM tended to be greater for ICW_{BIS} than for $eSMM_{BIA}$. Thus, we postulated that BIS may potentially offer a more discerning reflection of SMM estimation.

There were some limitations to this study. First, we only validated the methods among male athletes; therefore, sex-based differences could not be observed. The distributions of ICW and ECW in the body fluctuate with the menstrual cycle and should be examined in future studies involving female subjects. Additionally, the glycogen concentrations of muscles may be impacted by diet and exercise prior to evaluation. Our previous study showed an increase in the ICW baseline by 1.6% at 72 h after high-intensity glycogen depletion exercise with high-carbohydrate diets [50]. To address these factors, we prohibited exercise for 24 h, asked the participants to stay overnight at the facility the day before the measurements, provided them with a standard Japanese dinner, and limited their fluid intake to 500 mL or less after dinner. Additionally, we ensured a 10 h fast prior to the measurements. Moreover, the study participants were asked to avoid vigorous exercise 24 h before the measurements to ensure that impedance was not affected by muscle damage [51]; however, peak muscle damage appears 4 d after exercise. We restricted the activities of the participants only on the day before the measurements because most of them were regularly active athletes.

In conclusion, this study examined the association between SMM and the predictability of single-frequency BIA, multi-frequency BIA, and BIS as methods for estimating whole-body SMM , ECW , and ICW in active young men, including athletes, using stable dilution techniques as the reference methods. Importantly, notable correlations were identified. SMM and ICW exhibited significant correlations with single-frequency BIA (L^2/R_5 and L^2/R_{50}), multi-frequency BIA (L^2/R_{250-5}), and BIS (L^2/R_{ICW}). Specifically, the r of SMM and ICW were stronger with single-frequency BIA at L^2/R_{50} than that at L^2/R_5 . Hence, it may be deduced that using water compartment tracer data could substantiate the predictability of impedance methods and that there are reasonably robust associations between SMM and the impedance indices; therefore, we propose the feasibility of estimating SMM using impedance techniques. These results should be cross-validated among other samples to confirm whether impedance methods could be used for practical predictions of SMM . Bioimpedance technologies comprise a weak bioelectric current; therefore, they are minimally invasive, safe, user-friendly, and expeditious. Body composition assessments using bioimpedance are widely conducted at research institutes, medical institutions, gyms, medical examination centers, and even households. Further studies are warranted to ascertain or establish the most suitable estimation equations for specific population cohorts and determine whether it is technically

viable to use any impedance device and the same measurement protocol to obtain identical impedance values.

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Author contributions

The authors' responsibilities were as follows—HS, YY, and HT: planned the study; HS, EK, TY, TO, and HT: collected data; MS, EN, WJE, MH, KS, TY, JY, and YH: performed specimen and data analyses; HS, AU, KS, TY, and YY: conducted statistical analyses; HS: prepared illustrations; HS, EK, AU, and YY: drafted the manuscript; and all authors: read and approved the final manuscript.

Conflict of interest

WJE and MH are co-inventors on patents for the D3-creatine dilution method. However, they do not control the IP or derive any income from using this method. All other authors report no conflicts of interest.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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